DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM
August 2013
ALGEBRA (Ph.D. Version)

Instructions to the student

a. Answer all six questions; each will be assigned a grade from 0 to 10.

b. Use a different booklet for each question. Write the problem number and your **code number (not** your name) on the outside of the booklet.

c. Keep scratch work on separate pages in the same booklet.

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1. Let p be a prime and let G be a group of order p^k for some $k \geq 1$.
(a) Suppose G acts on a finite set S. Let $F = \{x \in S \mid gx = x \text{ for all } g \in G\}$. Show that $\#S \equiv \#F \pmod{p}$.
(b) Suppose $N \neq 1$ is a normal subgroup of G. Let Z be the center of G (so Z is the set of elements of G that commute with everything in G). Show that $N \cap Z \neq 1$.

2. Let T be a linear transformation of a finite-dimensional complex vector space V.
(a) Suppose that, for each subspace W of V such that $T(W) \subseteq W$, there is a subspace U of V (depending on W) such that $T(U) \subseteq U$ and $V = W \oplus U$. Show that T is diagonalizable.
(b) Suppose that T is diagonalizable. Let W be a subspace of V and suppose that $T(W) \subseteq W$. Show that there is a subspace U of V such that $T(U) \subseteq U$ and $V = W \oplus U$.

3. Let $R = \mathbb{C}[x,y]/(x^3, y^3)$, where (x^3, y^3) denotes the ideal of $\mathbf{C}[x,y]$ generated by x^3 and y^3.
(a) Find all prime ideals of R.
(b) Show that R has a unique maximal ideal.
(c) Find all units of R.

4. Let $A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ be an exact sequence of free modules (over some ring with 1).
(a) Show that $B \simeq \mathrm{Im}(f) \oplus C$ by giving an explicit map and showing that it is an isomorphism. (Do not quote results about projective modules and splitting of exact sequences.)
(b) Show that there exist module homomorphisms $v : B \rightarrow A$ and $w : C \rightarrow B$ such that $fv + wg$ is the identity map on B.

5. Let L/K be a finite Galois extension of fields and let $G = \mathrm{Gal}(L/K)$. Let $\alpha \in L$ with $\alpha \notin K$.
(a) Show that there exists $g \in G$ with $g(\alpha) \neq \alpha$.
(b) Show that there exists $h \in G$ with h of prime-power order such that $h(\alpha) \neq \alpha$. (That is, $h^{p^k} = 1$ for some prime p and some $k \geq 1$.)
(c) Suppose that every field F with $K \subsetneq F \subseteq L$ contains α. Show that G is cyclic of prime-power order.

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6. Let G be a finite group and let H be a subgroup of G. Let $\rho : G \to GL(V)$ be a finite-dimensional irreducible representation of G on a complex vector space V.

(a) Let $W \neq 0$ be a subspace of V such that $\rho(h)W = W$ for all $h \in H$. Show that

$$\sum_{g \in G} \rho(g)W = V.$$

(b) Suppose H is abelian. Show that $\dim(V) \leq [G : H]$.

(c) Show that if H is abelian then the number of distinct irreducible characters of G is greater than or equal to $|H|^2/|G|$.